Problem 00000003901: a genuine Frobenius quotient length

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Result. In the actual one-dimensional ring

$$R = \mathbb{F}_2[[X]], \qquad I = (X), \qquad q = 2^e,$$

the natural Frobenius quotient has actual R-module length

$$\ell_R\big(I/I^{[q]}\big) = q - 1 \qquad \text{for every } e \geq 0.$$

It is unbounded along e and is not $O(q^{1-1}) = O(1)$. In fact its normalization by q^{1-1} cannot tend to any finite real coefficient. This refutes the necessary proposed exponent $n - 1$ for the natural corrected quotient, without relying on finitely many sample values.

Printed quotient direction. The source writes $I^{[q]}/I$. A module quotient of ideals requires the denominator to be contained in the numerator. In our example $I^{[q]} \subseteq I$; Lean also proves $I \not\subseteq I^{[2]}$. Thus the printed direction is not even defined as an ordinary quotient already at $q = 2$. Our substantive counterexample addresses the natural correction $I/I^{[q]}$, as did the earlier review's intended length discussion. The direction error is disclosed rather than used as the only objection.

1. Actual ring, ideal and Frobenius power

The ring is the genuine formal power-series ring over `ZMod 2`, not an assigned object with a dimension label. Its discrete-valuation-ring structure, maximal ideal $I = (X)$ and actual Krull dimension 1 are proved using Mathlib's power-series, DVR and PID dimension theorems. The source quantifies over every ideal of an n-dimensional ring and imposes no polynomial-ring-only restriction, so this ring is within that domain.

For an ideal J in characteristic p, the standard p^e-th Frobenius power is generated by the p^e-th powers of its elements; see [1], Introduction, printed p. 1. Lean uses exactly the ideal map under the actual iterated Frobenius endomorphism and proves

$$I^{[2^e]} = \text{Ideal.map}(F^e)(I) = (X^{2^e}).$$

It also proves this ideal is contained in I. The quotient denominator is its actual inverse image under the inclusion $I \hookrightarrow R$, so the resulting type is a genuine quotient R-module of I.

The example is a one-variable monomial ideal in a power-series ring. Its exponent set is $\{1, 2, \ldots\}$ and its usual Newton polyhedron is $[1, \infty)$. No independently formalized Newton boundary-measure formula is needed or asserted: failure of the necessary exponent already precludes the full claimed exponent-and-coefficient statement.

2. A true linear equivalence computes the length

Multiplication by X is an actual R-linear bijection $R \to I$. Injectivity follows from $X \neq 0$ and the domain property; surjectivity is the literal divisibility characterization of membership in (X). For each $k \geq 0$, its inverse image of (X^{k+1}) is (X^k), by cancellation of X. Thus it induces the real R-linear equivalence

$$R/(X^k) \;\cong_R\; I/(X^{k+1}).$$

Mathlib's DVR quotient-length theorem gives $\ell_R(R/\mathfrak{m}^k) = k$. Rewriting $\mathfrak{m} = (X)$ and transporting *module length* through this linear equivalence yields

$$\ell_R(I/(X^{k+1})) = k.$$

For $q = 2^e > 0$, take $k = q - 1$. This proves the displayed all-e formula for $I/I^{[q]}$. In particular, we never replace this module with $R/I^{[q]}$, whose length would be q, nor rename a vector-space dimension as an R-module length.

3. The full asymptotic obstruction

Lean's `Module.length` takes values in the extended natural numbers. The proof first establishes the actual finite value $q-1$ and its inequality to infinity. Only then does it extract the real numerical length $L(e)$ using `toNat`, proving $L(e) = (2^e - 1)$.

Since $2^e - 1 \geq e$, for every real C and natural cutoff N one can choose

$$e = N + \lceil C \rceil_+ + 2 \geq N \quad \text{with} \quad C < L(e).$$

This is proved with all constants and cutoffs quantified. It contradicts any eventual real bound, and hence the genuine Mathlib statement $L = O(1)$ is false. As $n = \dim R = 1$, this is precisely the proposed q^{n-1} scale. The source's choice of leading coefficient cannot repair the exponent: for every finite real c, the actual normalized function $L(e)/(2^e)^{1-1}$ does not tend to c; a finite limit would eventually bound it by $c+1$, contradicting the same cofinal inequality.

The formal result is this failure of the proposed zero exponent and finite leading coefficient. We do not additionally claim to have formalized a general theory of leading exponents, the exact leading exponent 1, or Newton boundary measures. No finite numerical table substitutes for the asymptotic argument.

4. Earlier submission and verification

Public PR 285 already suggested a maximal-ideal Frobenius quotient as the obstruction, using a two-variable polynomial ring and the proposed length $q^2 - 1$. That exposed direction is attributed. The maintainer review rejected a Lean project containing only four numeric length values and comparisons: no actual ring, ideal, Frobenius power, quotient module or length was represented, and no all-q growth argument was proved.

This submission uses a simpler one-dimensional power-series DVR and formally closes every one of those bridges. The ring and Krull dimension, true ideal maps, real quotient submodules, actual linear equivalence, genuine `Module.length`, finite-value extraction and all-cutoff asymptotic contradiction are present in Lean. The change of example is explicit; it is not presented as a reproof of the old two-variable formula or as a proof about the reversed printed quotient.

All 32 named local declarations are audited, including the proof-bearing characteristic-two instance and linear equivalences. The package includes pinned Lean/Mathlib versions, a newly compiled root module, warning-as-error axiom audit, full fresh kernel replay and source-bound receipts. Full original bilingual source, LaTeX source and PDF are provided. There is no `sorry`, `native_decide` or additional axiom.

AI disclosure. Research, Lean proofs and documents were produced by a Codex worker with separate AI supervision of source semantics, proof scope and PDF layout. Organizer acceptance and human verification are not asserted.

References

[1] C. Huneke, M. Katzman, R. Y. Sharp and Y. Yao, *Frobenius test exponents for parameter ideals in generalized Cohen–Macaulay local rings*, J. Algebra **305**(1) (2006), 516–539. Author version, Introduction p. 1 (PDF page 2, after repository cover), White Rose repository; doi:10.1016/j.jalgebra.2006.06.036.